Web Scraper and OpenRefine

Data Collecting, Wrangling and Cleaning

CESTA

Workshop Materials

All materials are available in the GitHub repository

https://github.com/KGeorgii/CESTA_RS_2026/tree/main/OpenRefine

- Presentation
- Web Scraper guide
- Web Scraper practice setup, note
- Web Scraper practice setup, prerecorded video (YouTube)
- OpenRefine guide
- OpenRefine practice dataset

Part I
Web Scraping

Web Scraping

Web scraping is the automated process of extracting data from websites using software/code

How it works

1. Sends a request
2. Fetches the website (*downloading the page's raw HTML code*)
3. Parses the content (*reading the HTML structure*)
4. Extracts the data
5. Repeats that action
6. Saves the data

Web Scraping

Ethics

1. Check the terms of service
2. Check robots.txt*
3. Don't overload the server
4. Respect private and personal data
5. Don't scrape for unfair profit
6. Give credit

**filename used to implement the Robots Exclusion Protocol, a standard that websites use to indicate to visiting web crawlers and other web robots which portions of the website they can and cannot access. Websites have this file at website.com/robots.txt*

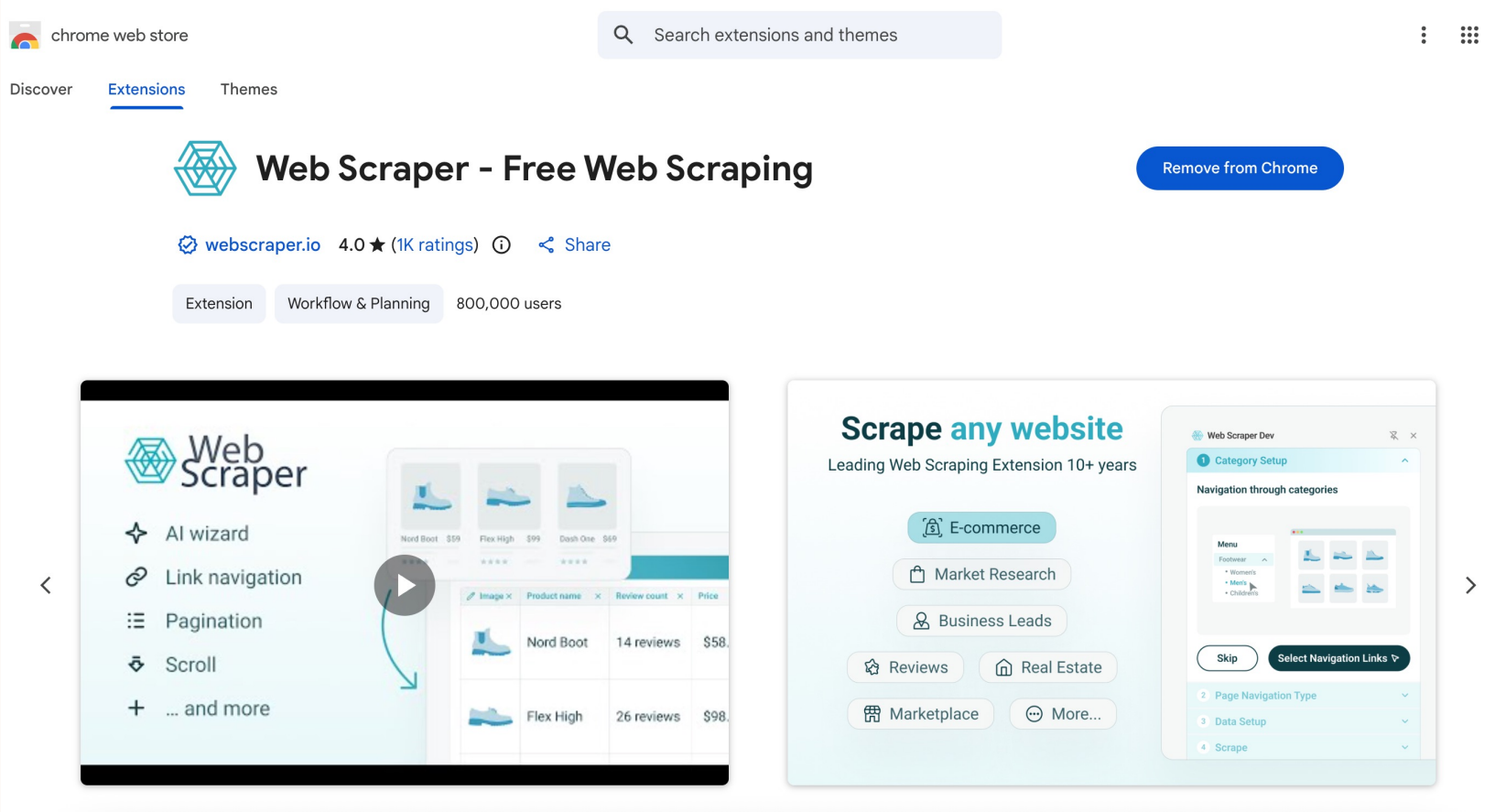
Web Scraping with Web Scraper

Web Scraper

Chrome extension

Key Features

- Zero coding required
- Pre-built recipes
- Data exports for further analysis
- Handles dynamic content and modern websites



Web Scraping with Web Scraper

Web Scraper

Official website: https://webscraper.io
Documentation: https://webscraper.io/documentation
Video tutorials: https://webscraper.io/tutorials

Web Scraping with Web Scraper

Web Scraper Installation

1. Open Google Chrome
2. Navigate to Chrome Web Store
3. Search for "Web Scraper"
4. Select the official Web Scraper extension by webscraper.io
5. Click Add to Chrome → Add Extension

Web Scraping with Web Scraper

Core Concepts

Sitemap — scraper's instruction from start to finish; it defines what pages to visit, what selectors to use, and how to save the data

Selector — points to the data you want; a rule that tells the scraper which part of the webpage to look at and extract data from

_root — the starting point of every sitemap. Represents the seed URL you entered when creating the sitemap. The first page the scraper lands on before any selectors run.
All top-level selectors set _root as their parent

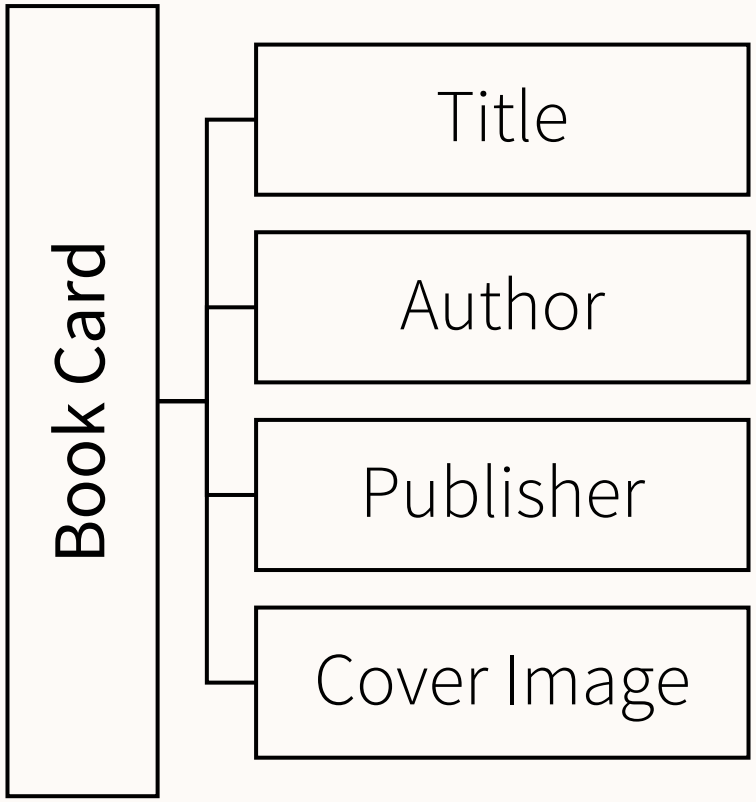
Web Scraping with Web Scraper

Parent-Child Hierarchy

Parent-Child Hierarchy — the foundation of structured data extraction
Parent Selector (wrapper) — defines the boundary of each data record
Child Selectors (data points) — extract specific information within each parent
Parent Selector = Row, Child Selectors = Columns

How It Works
Parent selector identifies repeating elements (products, articles, listings)
Child selectors extract data from within each parent boundary
Parent creates one row in your final dataset. Child selectors become columns in that dataset.

Web Scraping with Web Scraper



id	title	author	publisher	cover
Book_1				
Book_2				

Child selectors
Define columns
Extract data for each book

Parent selector
Opens every book card
Creates new rows

Parent-Child Hierarchy

example structure

Web Scraping with Web Scraper

Selector types

Selectors are used for different types of data extraction and interactions with the website

Can be divided into 3 groups:

- Data extraction selectors for data extraction
- Link selectors for site navigation
- Element selectors for element selection that separate multiple records

Web Scraping with Web Scraper

Selector types

Text Selector
Extracts: Plain text
Best For: Headlines, descriptions, prices, reviews

Image Selector
Extracts: Direct links to image files
Best For: Product photos, gallery images, profile pictures, etc.

Link Selector
Extracts: URLs and enables multi-page scraping
Best For: Navigation links, detail pages

Click Selector
Clicks something to reveal hidden content
Best For: Tabs, accordions, dropdowns, expandable sections

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Selector types

Table Selector
Purpose: Extracts structured data directly from HTML tables. Identifies the table element and automatically maps rows and columns into a structured dataset

Element Scroll Selector
Purpose: Automatically scrolls to load more content before extraction
Best For: Social media feeds, endless product listings

Pagination Selector
Purpose: Automatically navigates through multiple pages to extract data from the entire dataset. Detects common pagination patterns across different websites. Stops when reaching the last page or when no more content is available

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! Note

multiple Setting

Controls how many elements the scraper collects from a selector on a given page. Default false — grabs the first matching element only. Enable multiple to ensure all repeating elements are extracted. Best for parent selectors on listing pages, where the selector must catch every item on the page, like product cards.

Web Scraping with Web Scraper

! Note

Pagination as Parent

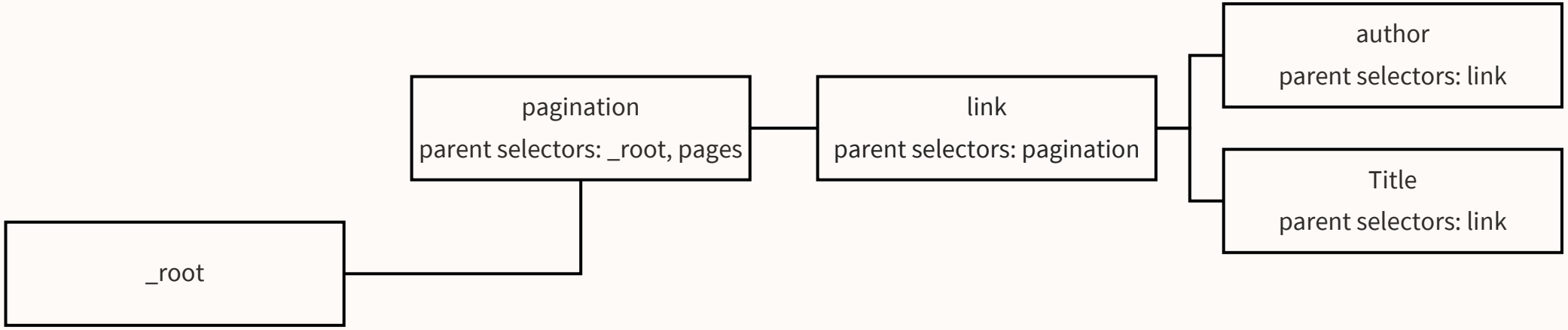
Pagination always references itself as a parent to keep looping through pages until the last one is reached.
Link selector almost always should be parented to pagination. This ensures painting links, product cards, or article links are collected from every page, not just page 1.

Web Scraping with Web Scraper

! Note

Pagination as Parent

structure



Web Scraping with Web Scraper

Sitemap Controls

Edit Metadata
Change the root link (start URL)
Rename the sitemap

Change Order
Rearrange the order of columns in your final dataset

Element Preview
Verify the selector is targeting the correct element on the page

Data Preview
Verify data extraction is working correctly
Check that parent-child relations are correct

Add Selector
Create a new selector to capture an additional data point

Delete Selector
Remove a selector entirely from the sitemap

Edit Selector
Modify an existing selector's type, rule, or label

Web Scraping with Web Scraper

Best Practices

- Identify repeating patterns on the target website
- Use descriptive names for easy management
- Plan your parent-child hierarchy before creating selectors
- Keep selectors simple
- Test on a small sample before full extraction
- Test across multiple pages to ensure consistency
- Verify selectors capture intended data

Web Scraping with Web Scraper

Basic Workflow

1. Access the Web Scraper extension in your browser
2. Create a new sitemap
3. Add selectors
4. Test your selectors
5. Run the scraper
6. Verify the results
7. Export the dataset

Practice
The MET Museum
European Paintings Collection

Part II
Data Wrangling and Cleaning

Data Wrangling and Cleaning

Key Rule

Data must be structured, consistent, and clean. Same format, same spelling, same structure across every record.

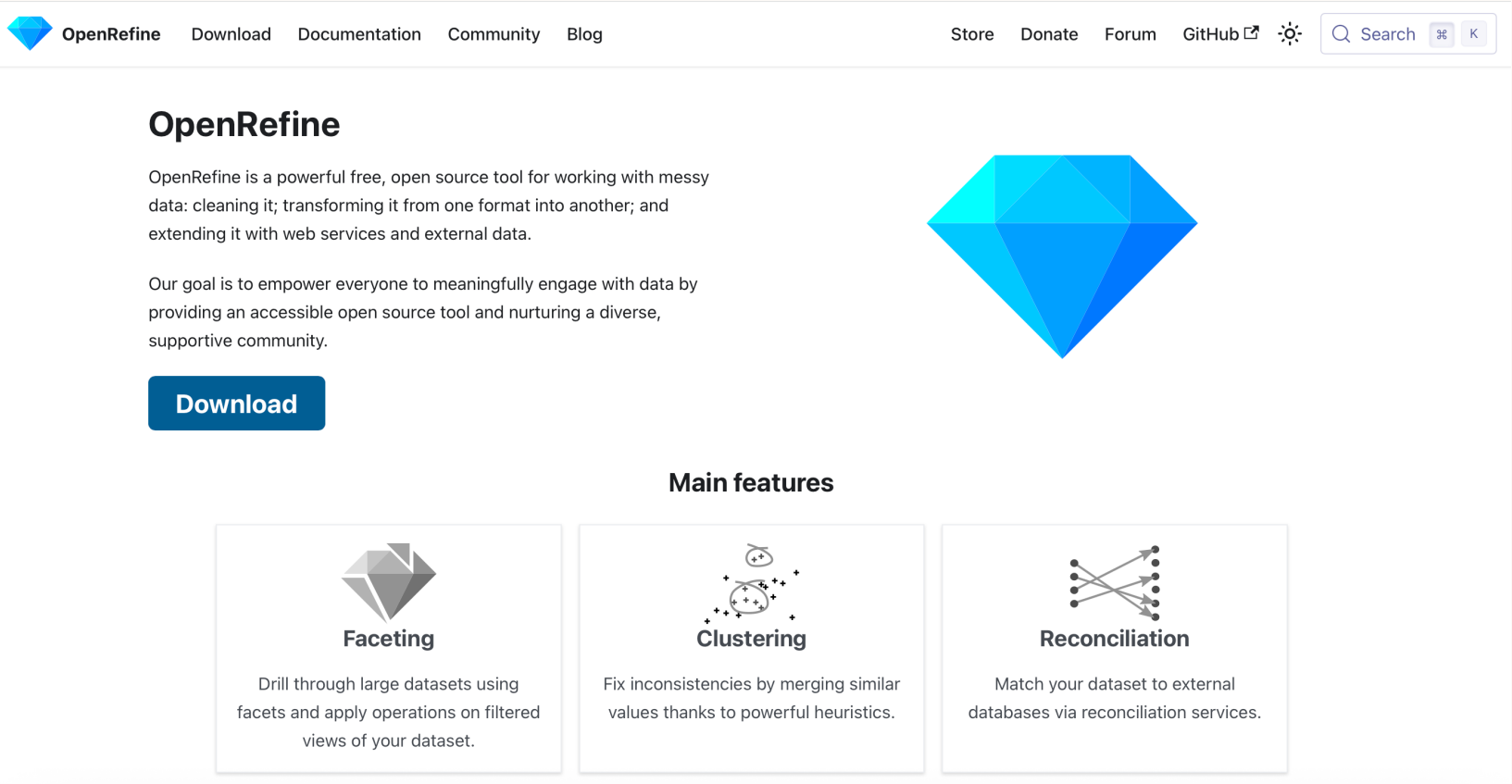
Inconsistencies can lead to unreliable statistics and wrong conclusions.

Data Wrangling and Cleaning. OpenRefine

OpenRefine

OpenRefine is a powerful tool for working with messy data: cleaning it, transforming it, and extending it.

- Free
- Open Source
- Runs locally on your device — your data never leaves your machine
- No connection needed



Data Wrangling and Cleaning. OpenRefine

OpenRefine

Official website: https://openrefine.org
User manual: https://openrefine.org/docs
Extensions: https://openrefine.org/extensions

Data Wrangling and Cleaning. OpenRefine

Metadata and Data. The Difference

When working with digital collections, it’s essential to understand what kind of data you’re working with. This distinction isn’t just technical — it defines the tools we are using and shapes how we analyze, visualize, and interpret materials.

What Is Metadata?
When you select a book in a library, you’re not reading the book itself — you’re reading its metadata. You check the author, the publisher, the year, maybe the number of pages — clues that help you decide whether the book is relevant or interesting. These details don’t contain the content of the book, but they provide essential context about it. This is metadata. *Data is the content of the book.*

Metadata is data about data — it describes, classifies, and provides structure for the content you’re working with.

Data Wrangling and Cleaning. OpenRefine

Data and Metadata. The Difference

Metadata

Data

Data Wrangling and Cleaning. OpenRefine

OpenRefine is a *metadata* cleaning tool.

It works on structured descriptive fields, not on raw content.

Data Wrangling and Cleaning. OpenRefine

OpenRefine Key Features

Faceting
Browse and filter data by unique values. Instantly see how many times each value appears — spot inconsistencies, typos, and outliers across the entire dataset at a glance.

Clustering
Groups similar values that likely mean the same thing — "New York", "new york", "N.Y." — and merges them in one click.

Mass Edit
Apply a single change to every matching cell at once. Rename a value, fix a typo, or standardize a format across hundreds of rows instantly.

Data Wrangling and Cleaning. OpenRefine

OpenRefine Key Features

Transformations
Apply built-in or custom functions to clean and reshape values — trim whitespace, change case, split or join fields, extract patterns using regex.

Undo / Redo History
Every operation is logged. Step back to any point in your cleaning process without losing work.

Export
Export cleaned data as CSV, Excel, JSON, and other formats — ready for further use or upload.

Data Wrangling and Cleaning. OpenRefine

OpenRefine Key Features

Extensions

There are multiple extensions available to extend the core functionality.
Key extensions include Wikidata (for reconciling and uploading data directly from and to Wikidata) and the RDF extension for linked data workflows.

Extensions are listed at: https://openrefine.org/extensions

Data Wrangling and Cleaning. OpenRefine

Basic Workflow

1. Import data (create a new project)
2. Facet to explore and identify issues
3. Cluster to group and merge inconsistent values
4. Mass edit to standardize remaining fields
5. Transform to reformat and reshape
6. Export clean dataset

Practice
The MET Museum
European Paintings Collection